

Report on Spiral Model

Abstract:

This report presents a case study that explores the implementation of the Spiral model in Enterprise Risk Management Project. The Spiral Model is one of the most important Software Development Life Cycle models, which provides support for Risk Handling. The case study demonstrates how the Spiral model was applied to achieve successful results in a real-world scenario. It highlights the challenges, benefits, and lessons learned during the project's execution. The Spiral Model is a **Software Development Life Cycle** model that provides a systematic and iterative approach to software development. It is based on the idea of a spiral,

Introduction:

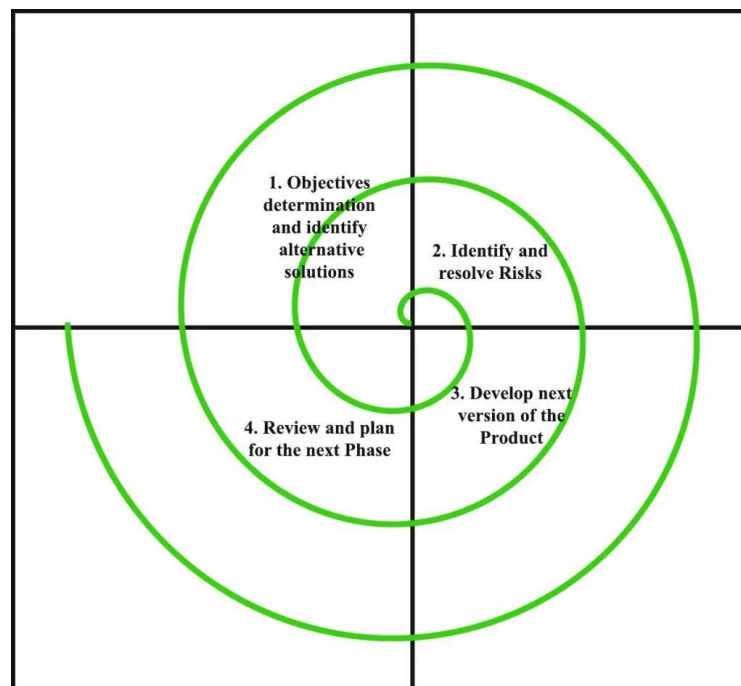
The Spiral Model is one of the most important Software Development Life Cycle models, which provides support for Risk Handling. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. Each loop of the spiral is called a Phase of the software development process. The Spiral Model is a Software Development Life Cycle model that provides a systematic and iterative approach to software development. It is based on the idea of a spiral, with each iteration of the spiral representing a complete software development cycle, from requirements gathering and analysis to design, implementation, testing, and maintenance. The Spiral Model is a risk-driven model, meaning that the focus is on managing risk through multiple iterations of the software development process. It consists of the following phases:

- **Planning:** The first phase of the Spiral Model is the planning phase, where the scope of the project is determined and a plan is created for the next iteration of the spiral.
- **Risk Analysis:** In the risk analysis phase, the risks associated with the project are identified and evaluated.
- **Engineering:** In the engineering phase, the software is developed based on the requirements gathered in the previous iteration.
- **Evaluation:** In the evaluation phase, the software is evaluated to determine if it meets the customer's requirements and if it is of high quality.
- **Planning:** The next iteration of the spiral begins with a new planning phase, based on the results of the evaluation.

The Spiral Model is often used for complex and large software development projects, as it allows for a more flexible and adaptable approach to software development. It is also well-suited to projects with significant uncertainty or high levels of risk.

The Radius of the spiral at any point represents the expenses(cost) of the project so far, and the angular dimension represents the progress made so far in the current phase. Each phase

of the Spiral Model is divided into four quadrants as shown in the above figure. The functions of these four quadrants are discussed below-



- **Objectives determination and identify alternative solutions:** Requirements are gathered from the customers and the objectives are identified, elaborated, and analysed at the start of every phase. Then alternative solutions possible for the phase are proposed in this quadrant.
- **Identify and resolve Risks:** During the second quadrant, all the possible solutions are evaluated to select the best possible solution. Then the risks associated with that solution are identified and the risks are resolved using the best possible strategy. At the end of this quadrant, the Prototype is built for the best possible solution.
- **Develop the next version of the Product:** During the third quadrant, the identified features are developed and verified through testing. At the end of the third quadrant, the next version of the software is available.
- **Review and plan for the next Phase:** In the fourth quadrant, the Customers evaluate the so-far developed version of the software. In the end, planning for the next phase is start.

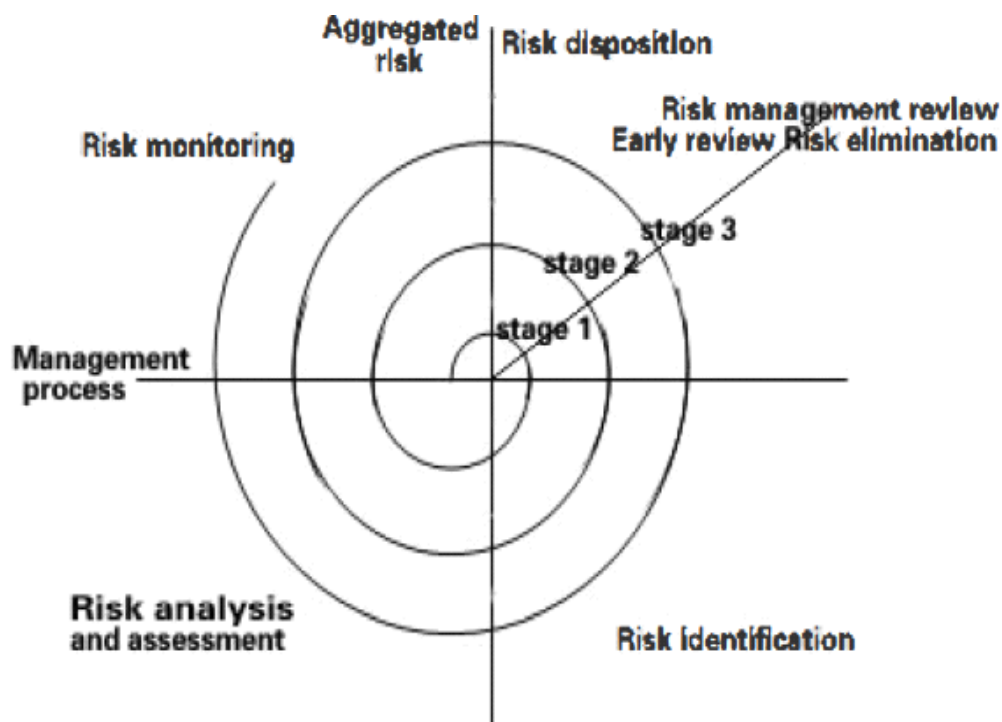
Problem Statement:

In today's fast-paced economy and digital world, businesses are growing quickly and facing more risks than ever before. These risks can harm a company's success. To protect themselves, companies need a good plan for dealing with risks. This study looks at how companies manage these risks using a special method called the Spiral Model. The question is: Can using the Spiral Model make risk management better for companies and help them

make more money? By studying this question, we are going to find out solution using the Spiral Model which can make companies safer and more successful.

Implementation of Spiral model on given problem :

Different enterprises have varying risk management goals based on their unique strategies. For instance, some aim to decrease the chances or impact of potential losses, while others, like high-tech companies with substantial investments and risks, prioritize increasing profit probability and amount. Unlike traditional risk management, enterprise risk management using the spiral model involves ongoing risk assessments at short intervals, with a repeated iterative process until all major risks are identified. This model breaks down enterprise management into stages of risk identification, analysis, monitoring, and disposition. After completing each management stage, a risk assessment is conducted to eliminate various risks before moving on to the next stage. The emphasis on risk analysis in the spiral model helps management personnel understand risks in each developmental stage and respond effectively. The review of risk management using this model aids enterprises in predicting future development, enabling them to devise skilled plans and attain goals in competitive markets by anticipating and addressing potential risks through predictive analysis. This approach involves considering likely risks, making informed predictions about the enterprise's future, and mitigating uncertainties during the planning and decision-making process.



Result:

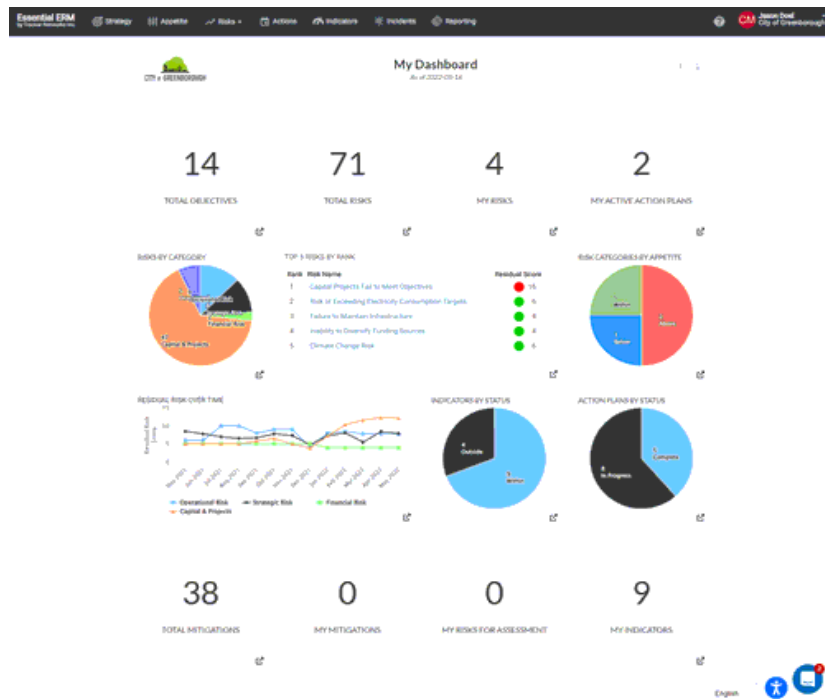


Fig. 1 Modular Components

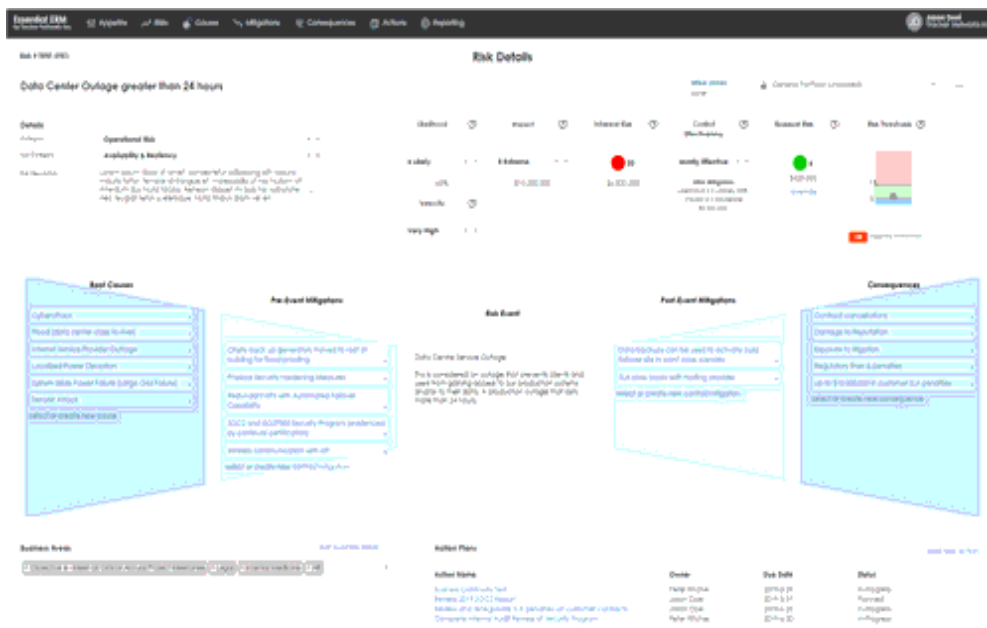


Fig. 2 Integrated Risk Bow Tie

Conclusion:

Risk management is an ever-evolving process that follows a dynamic cycle, much like a spiral model. This approach to managing risks brings about a more organized and well-thought-out way of running businesses. By effectively handling risks, companies can make their operations more logical and scientific, resulting in fewer risks overall. This, in turn, creates a safer environment for business activities, ensuring that goals are met smoothly and profits are maximized. The adoption of the spiral model in enterprise management holds great importance, both in theory and practice. It serves as a powerful tool for elevating the level of management, strengthening safety measures, and ultimately generating economic benefits. This approach proves to be valuable for businesses aiming to enhance their management practices, tighten safety controls, and secure financial gains.

The Application of Spiral Model in Enterprise Risk Management

Zhongwen Chen

Personnel department, Huanggang Normal University

Huanggang, China

chenzhongwen@hgnu.edu.cn

Abstract—The systematic risk management is of great importance in increasing enterprises' interests. With rapid development of economy and informatization and the expansion of enterprise, enterprises are faced with more and more risks. Therefore, it is vital and urgent for enterprises to establish risk management system. The paper analyzes the process of enterprise risk management based on the theory and methods of spiral model, and proposes that the Spiral Model can improve risk management level in enterprises and create economic benefits. (Abstract)

Keywords- risk; risk management; Spiral Model; risk identification; risk review (key words)

I. INTRODUCTION

The study of risk started in the early 20th century. In 1960s, J. G. March & H. A. Simon developed the management and decision theories with risk decision included. Risk problem has become one of the hot topics in the international academic community and has brought in worldwide upsurge of enterprise risk management since 1980s. The rise of the high-tech industry and the economic globalization lead to the growth of uncertainties. The condition of incomplete and asymmetric market information causes the increase of risks. Therefore, we have to consider the risk factor, enhance the risk consciousness, promote risk management and strive to prevent risk while capturing opportunities. For a country, it must guard against financial risk; for enterprises, market risk, operating risk, inventory risk, investment risk and technical risk, etc.

Risk is the possibility of loss. It is inevitable for an economic body in the activity of market economy and it is always uncertain but objectively existent^[1]. Risk losses, to the largest extent, might bring the crash of economic body. In the information age, the fundamental changes of economic environment, such as the global trade liberalization, the competition from the shorter product lifecycle, the consumers' demand for the appreciating currencies and the rapid development of information technology, etc., make information technology related risks loom larger. No risk management unreasonably concerns about the minor risks but pays most or all attention to the comparatively higher risk. The risk management is not without cost, which is esp. considerable when high-level personnel, high-quality equipments and high-precision information and data are used in management, and the cost is required to be less than or at least equal to the benefit brought about by the risk management.

In risk management, we can not make up the loss with benefit but should strengthen monitoring and predicting and risk review. Based on prevention, the risk can be reduced to a minimum, with its loss and impact diminished thereby. The aim of analyzing the forming factors of the risk, forecasting the risk and taking actively preventive measures is to avoid, transfer, detract, mitigate risks and prepare for assuming risks. Risk management is an indispensable part of the future management..

II. RISK MANAGEMENT

Risk management is involved in every industry. Enterprise risk management refers to the process of identifying, analyzing and assessing all risk factors possibly encountered by risk management departments to maximize security with the lowest cost. Generally, risk management is responsible for the risks in production activity. In the deep sense, it refers to how the subject, by means of risk identification, risk quantification, risk assessment, programs, controls and monitors risks and therefore more readily responds to the threats and achieves the target. The subject of the risk management is the management personnel, and the object risks or uncertainties in producing and operating. The large and complex manufacturing is required to set up specialized management sections and risk supervisors.

Risk management is a process consisting of risk identification, risk quantification, risk review, risk control and risk monitoring, which is to guarantee the production through the functions of planning, organizing, conducting and controlling and by applying overall scientific approaches. Risk management techniques, which count on optimization not sophistication, must be economical and in accordance with risk-cost-benefit relation. In every stage of cyclical risk management, a proper method of risk management must be adjusted to the ever-changing risk circumstances so as to achieve a dynamic total life-cycle risk management.

III. THE PROCESS OF RISK MANAGEMENT

The important part of risk management is risk awareness which is varied with organizations and individuals. PMI in America describes the risk management process as 6 sections: risk planning, risk identification, risk qualitative analysis, risk quantitative analysis, risk response planning, risk monitoring and controlling. However, SEI in America divides it into 6 parts^[2]: risk identification, risk analysis, risk planning, risk tracking, risk controlling and risk communication. In China, Bixing and Zhaili divide it into 4 stages^[3]: risk identification, risk analysis and assessment,

risk disposition, risk monitoring, and give a summary about each of them. The above divisions are similar aside from some differences.

Risk management, which is dynamic and must be implemented in the whole cycle of an enterprise, mainly includes risk identification, risk assessment, risk disposition and risk monitoring.

A. Risk identification

Risk identification, the first step of risk management, refers to identifying all the possible or potential risk sources and risk factors in the implementation process, defining and classifying their features and evaluating the nature of the risk. The purpose of risk identification is to reduce the uncertainties of system, i.e., discovering the main causing factors of risks and qualitatively evaluating the impact of risks. Two points should be made clear: to determine where the risks come from (the source) and classify them^[4]; to give an initial quantification toward the source of the risks. Risk identification is the basis of risk management, is a persistent and repetitive task. Any change of circumstances will probably lead to the change of the original risk and the birth of new risk. The identification of risk not only needs the judgment through perceptual knowledge, but more importantly depends on the analysis, summarization and classification of the various objective statistics and risk records, and therefore discovers the characteristics and rules of any risk. The common methods of risk identification are: expert survey (including brainstorming, Delphi method, interviewing, and questionnaire), scenario analysis and fault tree analysis, etc. Fault tree analysis (FTA) will be briefly introduced because of its wide application. It is powerful when analyzing the reliability and security of large complicated systems^[5]. We use FTA to analyze and reveal all the possible causes (sub-class events) of some unexpected event (extreme event). The logic relation between sub-class events and extreme event can be demonstrated as a reversed trees with the root of extreme event, hence the name FTA.

B. Risk assessment

Risk assessment follows risk identification. Both qualitative and quantitative analysis methods are used to assess the identified risk probability, scope, degree, range ability, distribution, duration and frequency so as to discover the main risk sources and factors affecting security and to determine the risk regions, risk sequence and the acceptable risk levels. Both the risk loss and its probability should be put into consideration to measure the risk gravity when making risk assessment. The purpose of risk assessment is to transfer various data into supportive information for decision maker, then evaluate risk consequences and finally determine the sequence of risk gravity. Risk assessment and risk decision criteria must be fixed before the impact of risk is assessed from the perspective of decision-making and the scale of the impact of risk on decision criterion is computed, hence receiving risk or not can be determined or the methods of controlling risks can be chosen to reduce or transfer risks. The relativity of risk loss must be noticed when analyzing and assessing the severity of risk loss, i.e., not only the

absolute amount of loss but also the capacity of the possible loss of an enterprise must be correctly evaluated. In the process of defining loss severity, every risk and various possible losses---direct and indirect, tangible and intangible---caused by all the risks, together with the overall impact of the losses on enterprise must be put into consideration. The risk impact is closely related with the time when loss takes place, duration and frequency, which are crucial for safe production.

C. Risk disposition

Risk disposition refers to putting forward suggestions and methods to deal with the risks. The overall consideration of risk probability, loss severity and other factors by risk identification and assessment enables to fix any possible risk and its severity which can be compared with accepted safety index so as to determine the hazard level and hence decide what measure to take and to what extent. Many effective strategies to deal with risks can be obtained from the changing of the nature of risk consequences, of the risk probability and of the extent of risk consequences.

D. Risk monitoring

Risk monitoring is to supervise and control the whole process of identifying, assessing and disposing the risk so that the risk management can be guaranteed to achieve its goal. To monitor risk is in fact to supervise the progress and condition of production---change of circumstances, the aim of which is to check whether the effect of the strategies and measures of risk management is the same with the prediction or not; to find a chance to improve and refine the risk control planning so as to obtain feedback and make the future decision more practical. In the process of risk monitoring, it is required to discover those new risks or the risks pre-established strategies and measures fail to control or the risks with its nature changing through time, then respond in time and once again identify, assess and dispose those risks according to their degree of influence on production while formulating criterion for success or failure of risk disposition. Risk monitoring, which is lack of a set of accepted techniques, must focus on the basic problems of risks that are of complexity, variability, unexpectedness, being ahead of time, formulate scientific criterion, adopt systematic management methods and establish an effective early warning system of risks and emergency planning to operate efficiently. Also, risk monitoring is a continuous process with its task being the overall tracking and assessing the disposing of the risks according to the measurement criterion formulated in the whole process of risk management. Effective risk monitoring can show the abnormal points in risk disposition and identify which risks are becoming actual problems, which enables the management to have plenty of time to take corrective measures. Meanwhile, it is crucial for risk monitoring to establish a management index system so that it can clearly and simply provide accurate, timely and relevant information about risks. The main methods of risk monitoring include inspection, monitoring list, risk report, etc.

IV. SPIRAL MODEL AND RISK MANAGEMENT

A. Spiral model

In 1988, Barry Boehm came up with a new model in software engineering, called spiral model, which is very suitable for large and complex system development because it combined the advantages of waterfall model and rapid prototyping model together and laid emphasis on the risk analysis neglected by other models.

Spiral model carries out system development by a cyclical way, which could develop a lot of intermediate revisions. Project manager can use it to demonstrate some notions in the early stage for the clients. The spiral model, which centers on the evolutionary development mode, is rapid prototyping method plus using waterfall model in every stage of the project. Within each cycle of the model are 4 stages: requirements definition, risk analysis, engineering fulfillment and assessment, which form an iterative process. Software development will be optimized once the process iterates. The software process with the spiral model adopted can be demonstrated as the following Figure 4.1:

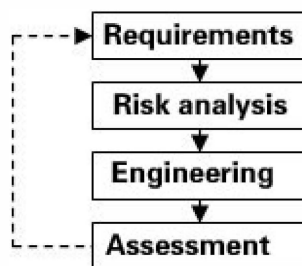


Figure 1. The software process of spiral model

The fulfillment of spiral model is to introduce at first a process of risk identification, risk analysis and risk control before every developmental stage, and the model breaks down the software project into smaller projects, each of which marks one or more main risks till all the main risks are identified.

Spiral model emphasize on risk analysis, which makes the developers and customers know about the risks appearing in each evolutionary stage and respond accordingly. Therefore, it is suitable for the large, complex and high-risk systems, for them risks are the potential, detrimental and unelectable factors in software development and could damage the development to different degrees and thus affect the quality of the software products. The purpose of reducing software risks is to identify and analyze risks in time and then decide to take proper measures to eliminate or decrease the damage of the risks.

B. The Application of Spiral Model in Enterprise Management

Enterprises have different risk management objectives because of their different strategies. For instance, some enterprises aim to reduce the risk loss or the likeliness of risk loss, and some, such as the high-tech enterprises characterized by high investment, high risk and high return

focus on the increase of the probability and quantity of profit. The difference between enterprise risk management with spiral model and traditional risk management is that there exists an overall assessment stage of risk management in every short period as well as a repetitive iteration process till all the main risks are identified. Spiral model divides each process of enterprise management into in-process risk identification, risk analysis and assessment, risk monitoring and risk disposition. A risk assessment is required for the previous process after the accomplishment of each process management to eliminate all kinds of risks and prepare for the next process management and then a repetitive iteration process is introduced to minimize the risks until the enterprise management competes. Spiral model emphasize on risk analysis, which makes the management personnel know about the risks appearing in each evolutionary stage and respond accordingly, which can be shown by the risk management scheme with spiral model in Figure 4.2:

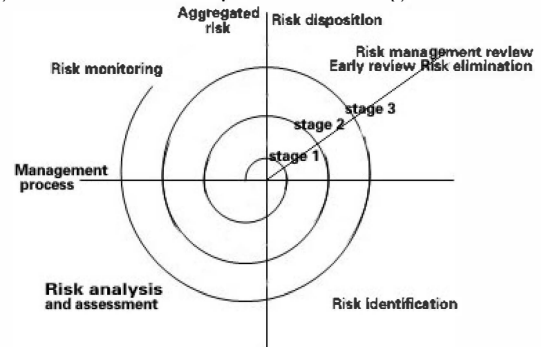


Figure 2. Spiral Model in Risk Management

The risk management review of spiral model can help the enterprises to predict their future development. An enterprise can work out plans and achieve goals skillfully and freely in the fierce market competition as long as it can infer and predict the remote and unknown by the present; it can keep persistent competitive advantage as long as it takes preventive measures beforehand. Predictive analysis is to take all the likely risks into consideration, make scientific prediction toward the future of the enterprise and eliminate all the risks and uncertainties arising in the process of planning and decision-making.

V. CONCLUSION

Risk management is a dynamic life cycle process in which spiral model is adopted. The operation of risk management can promote the scientific and rational management in enterprises and reduce their risks so as to provide a secure business environment, ensure the smooth accomplishment of business objective and increase their profits. It is significant, either theoretically or practically, to adopt the spiral model in enterprise management in terms of enhancing management level, strengthening safety control and creating economic benefits.

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